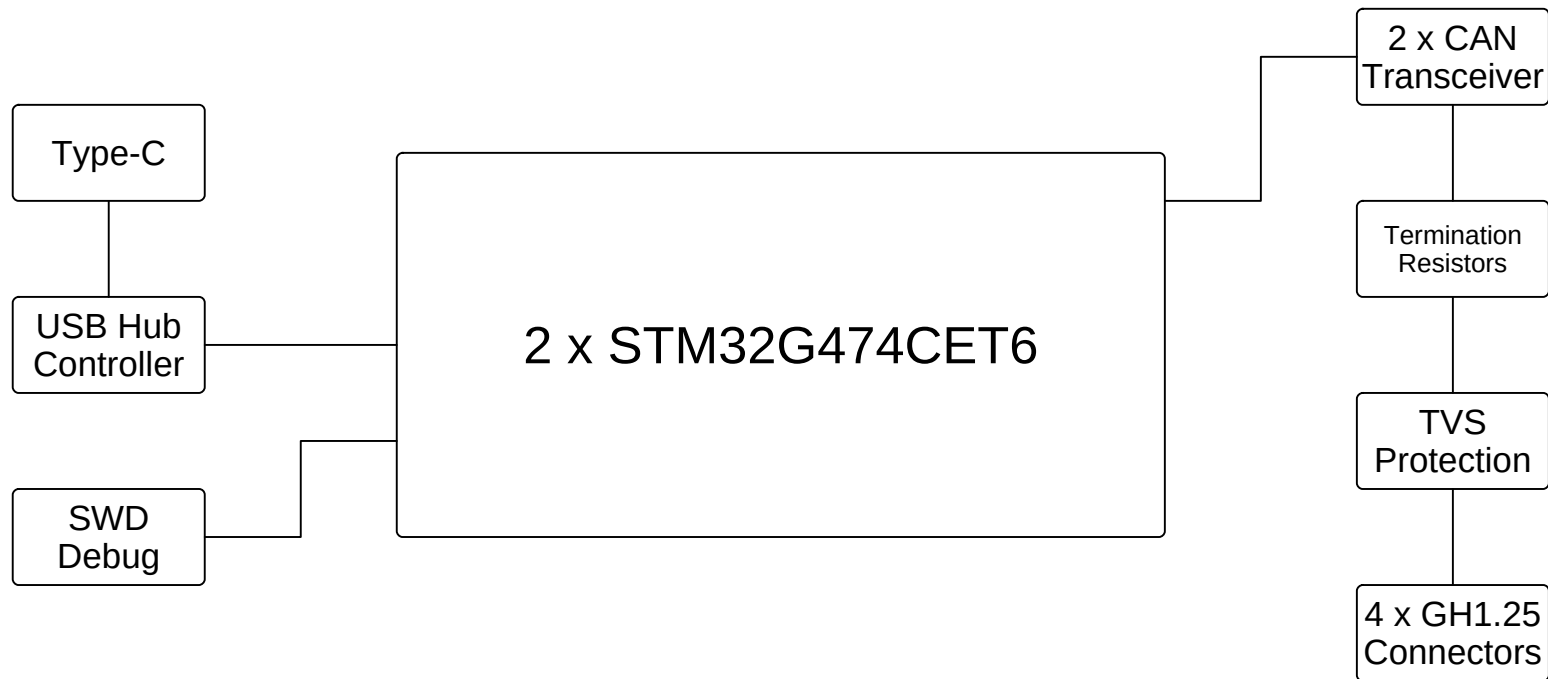
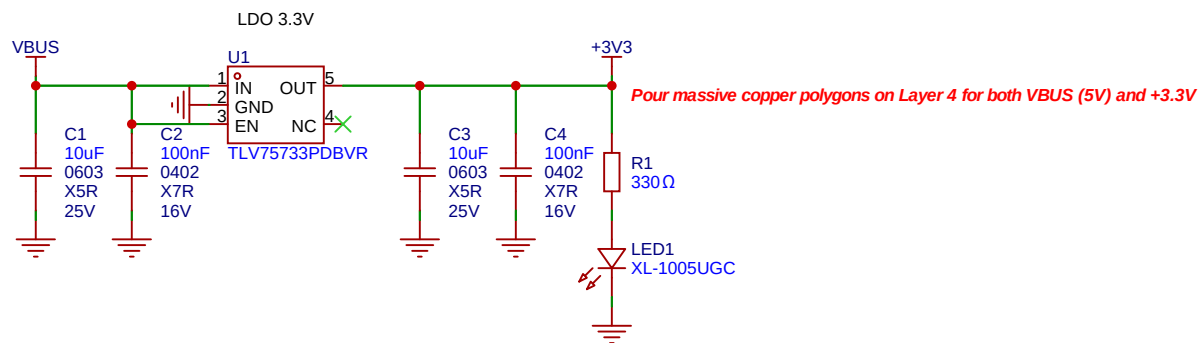


Date	Version Number	Details
July 25th 2026	V1.0	

Schematic	Schematic		Create at	2026-07-25
			Update at	2026-08-08
Board	Board_1_0		Page	Version
Drawn	Hieu N.	USB_CAN_Transceiver_Board		
Reviewed	Yigitler A.			
		Version	Size	Page 1 Total 8
		V0.0	A4	VNX Robotics



Schematic	Schematic			Create at	2026-07-25
Board	Board_1_0			Update at	2026-08-08
Drawn	Hieu N.	USB_CAN_Transceiver_Board			
Reviewed	Yigitler A.				
		Version	Size	Page 2 Total 8	
VN X ROBOTICS		V0.0	A4	VN X Robotics	



Component	Current Draw (Approx.)
FE1.1S-BSOP28BCN	70 mA @ 5V
STM32G474CET6 (x2)	35 mA each (running at 170 MHz) @ 3.3V
Can Transceiver Logic	20 mA @ 3.3V
LEDs	10mA @ 3.3V
Can Transceiver	@ 5V

Schematic	Schematic		Create at	2026-07-26
Board	Board_1_0		Update at	2026-08-09
Drawn	Hieu N.	USB_CAN_Transceiver_Board		
Reviewed	Yigitler A.			
		Version	Size	Page 3 Total 8
		V0.0	A4	VNX Robotics

A

B

C

D

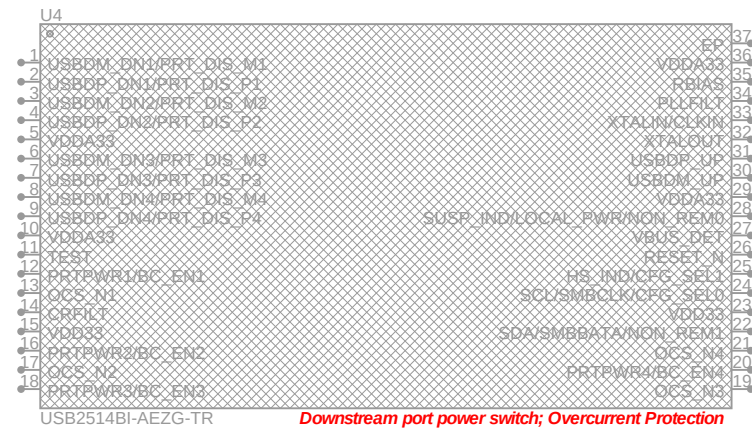
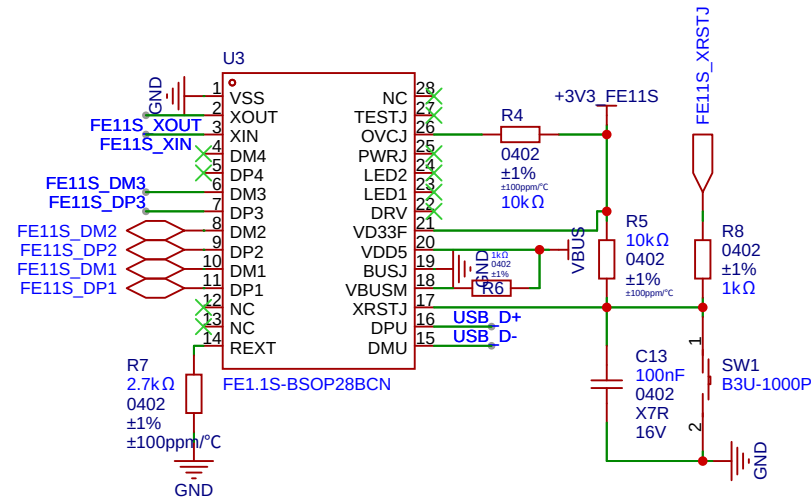
A

B

C

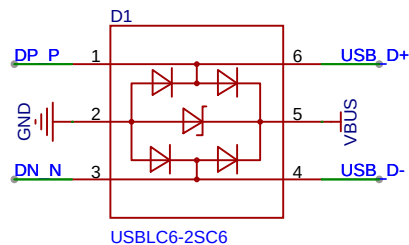
D

Place 5.1 kOhm res close to type-C
USB 2.0 (90-ohm Differential)

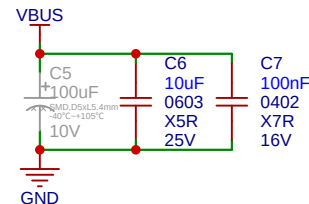


Downstream port power switch; Overcurrent Protection

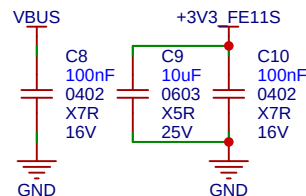
USB 2.0 Upstream ESD protection



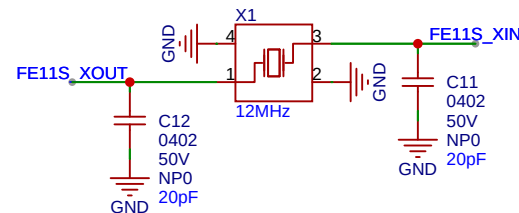
VBUS Filtering



USB Hub Power Filter

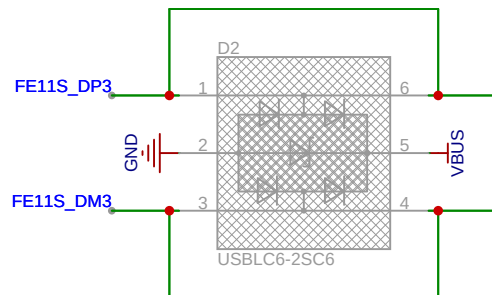


USB Hub Crystal Oscillator

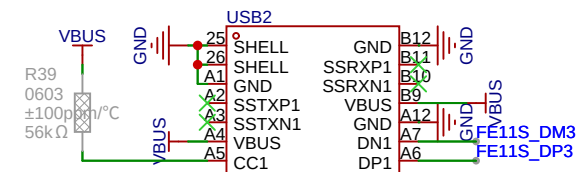


Traces from XIN and XOUT to the crystal pads symmetrically with matched lengths and under 10 mm.

USB 2.0 Downstream ESD protection



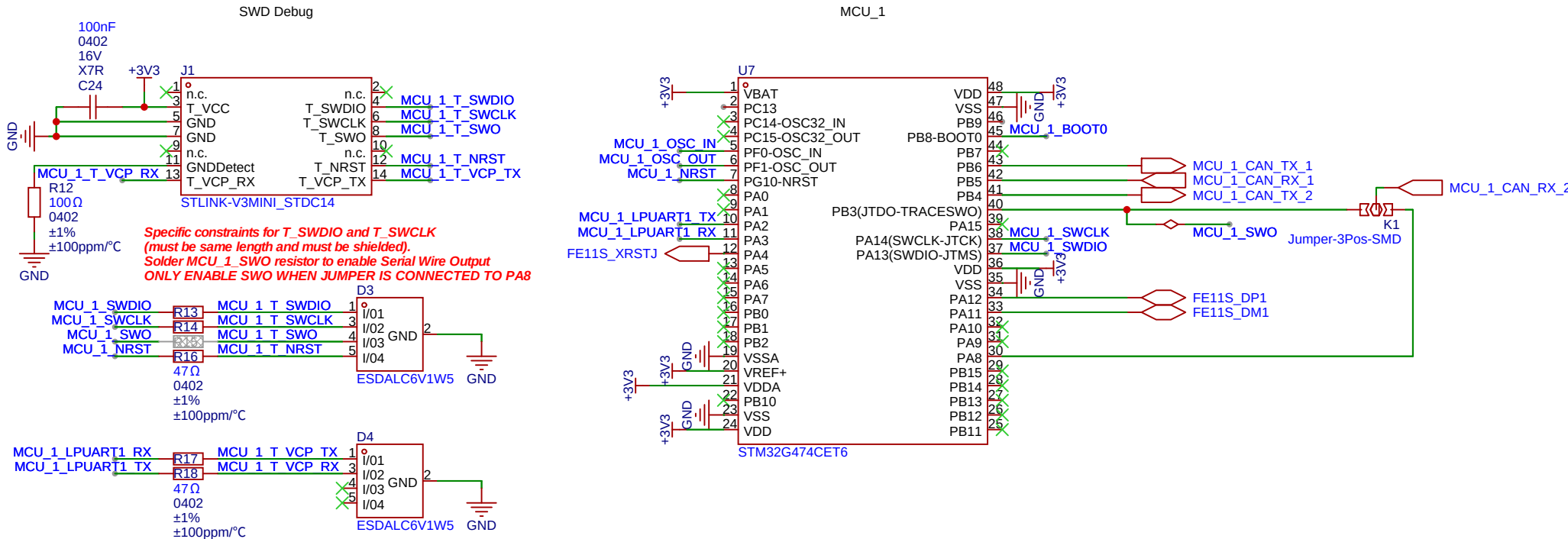
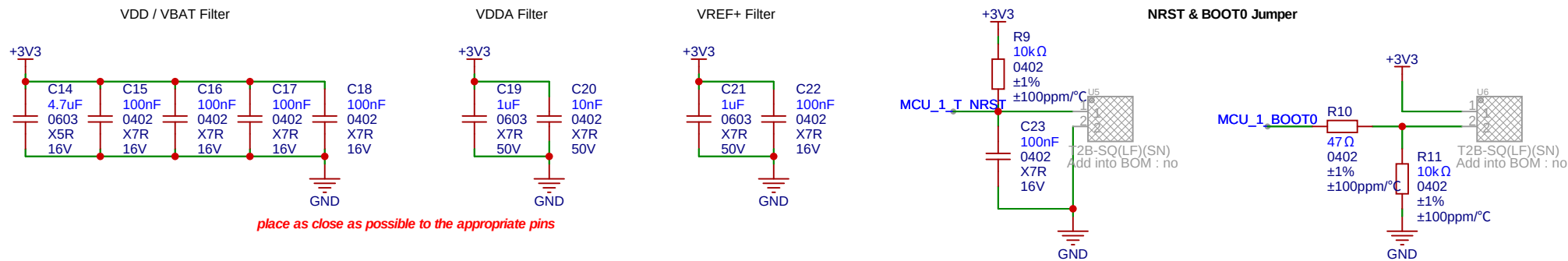
USB Type-C Downstream



U261-121N-4BS2S
Place 56 kOhm res close to type-C
USB 2.0 (90-ohm Differential)

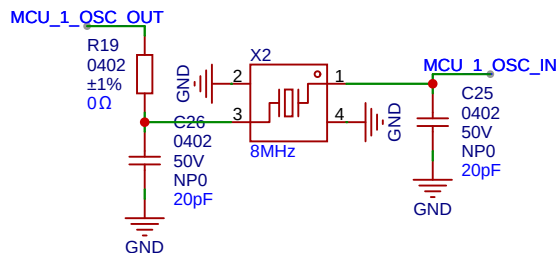
Schematic		Schematic		Create at	2026-07-25
Board		Board_1_0		Update at	2026-08-09
Drawn		Hieu N.		Page	USB
Reviewed		Yigitler A.		USB_CAN_Transceiver_Board	
		Version	Size	Page 4 Total 8	
		V0.0	A4	VNX Robotics	



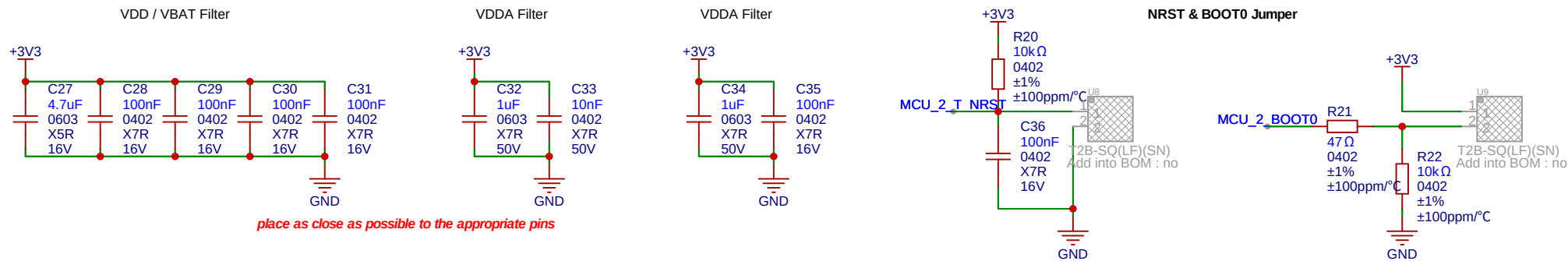


WARNING: Omitting SWO (PB3) disables zero-latency ITM/SWV printf logging and real-time variable tracing (Live Data Trace). All debug output must rely on LPUART1, which can introduce CPU blocking and timing jitter.

MCU 1 Crystal Oscillator

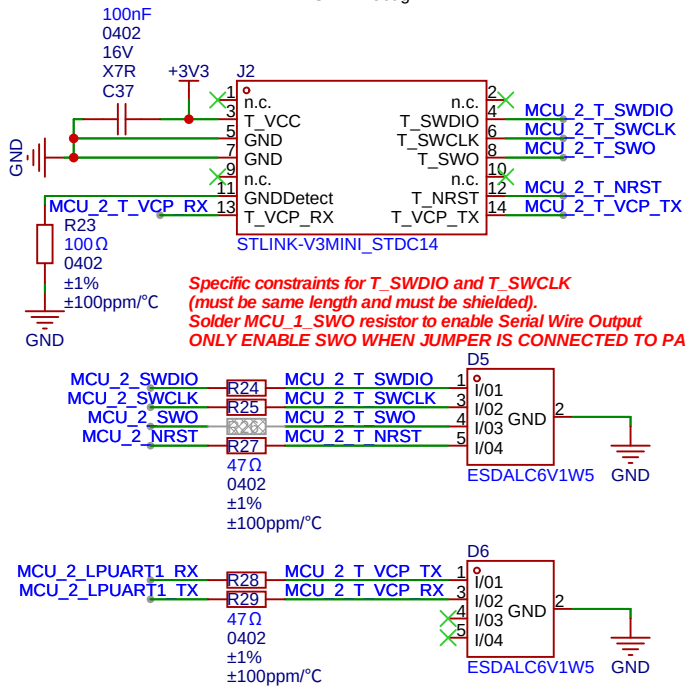


Schematic	Schematic	Create at	2026-07-26
		Update at	2026-08-09
Board	Board_1_0	Page	MCU_1
Drawn	Hieu N.	USB_CAN_Transceiver_Board	
Reviewed	Yigitler A.		
		Version	Size
		V0.0	A4
		Page 5 Total 8	
		VNX Robotics	



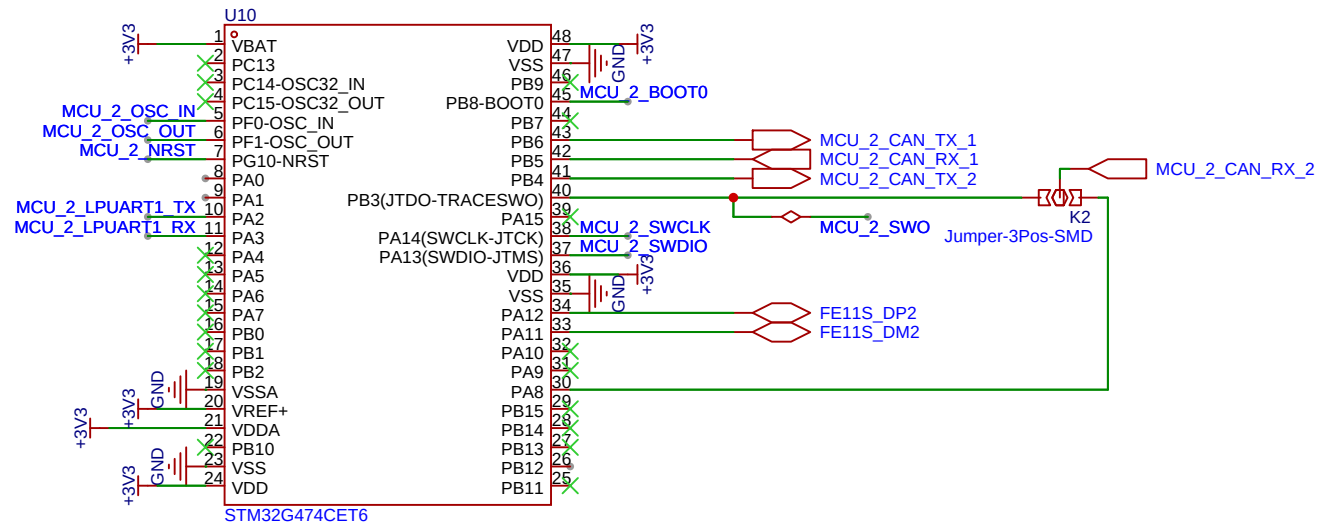
place as close as possible to the appropriate pins

SWD Debug



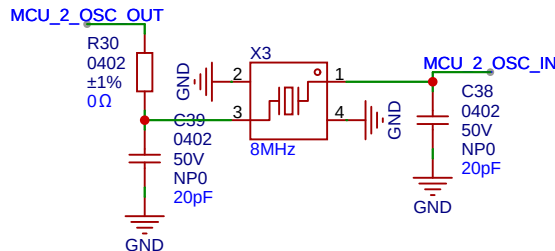
Specific constraints for T_SWDIO and T_SWCLK
(must be same length and must be shielded).
Solder MCU_1_SWO resistor to enable Serial Wire Output
ONLY ENABLE SWO WHEN JUMPER IS CONNECTED TO PA8

MCU_2

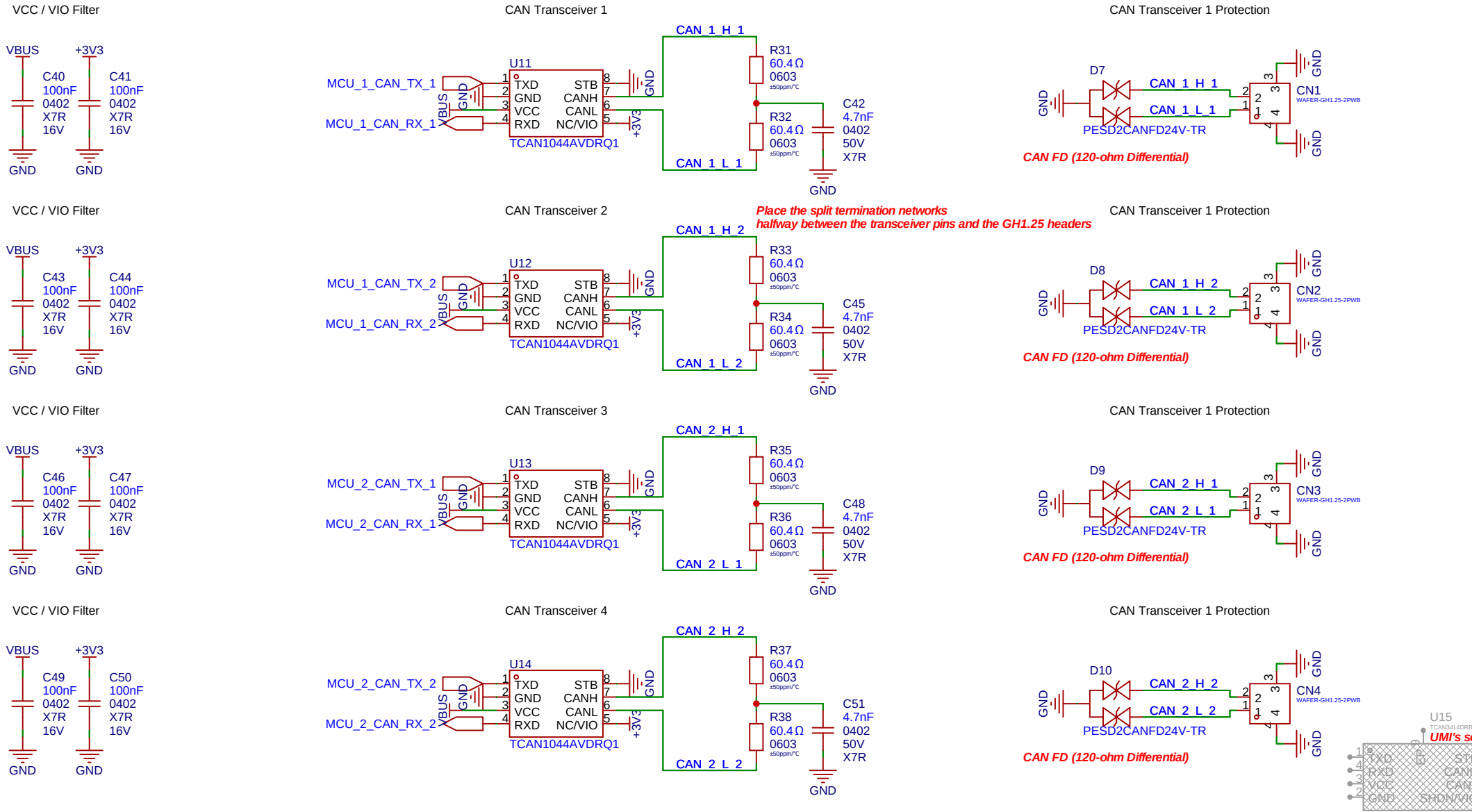


WARNING: Omitting SWO (PB3) disables zero-latency ITM/SWV printf logging and real-time variable tracing (Live Data Trace).
All debug output must rely on LPUART1, which can introduce CPU blocking and timing jitter.

MCU 1 Crystal Oscillator



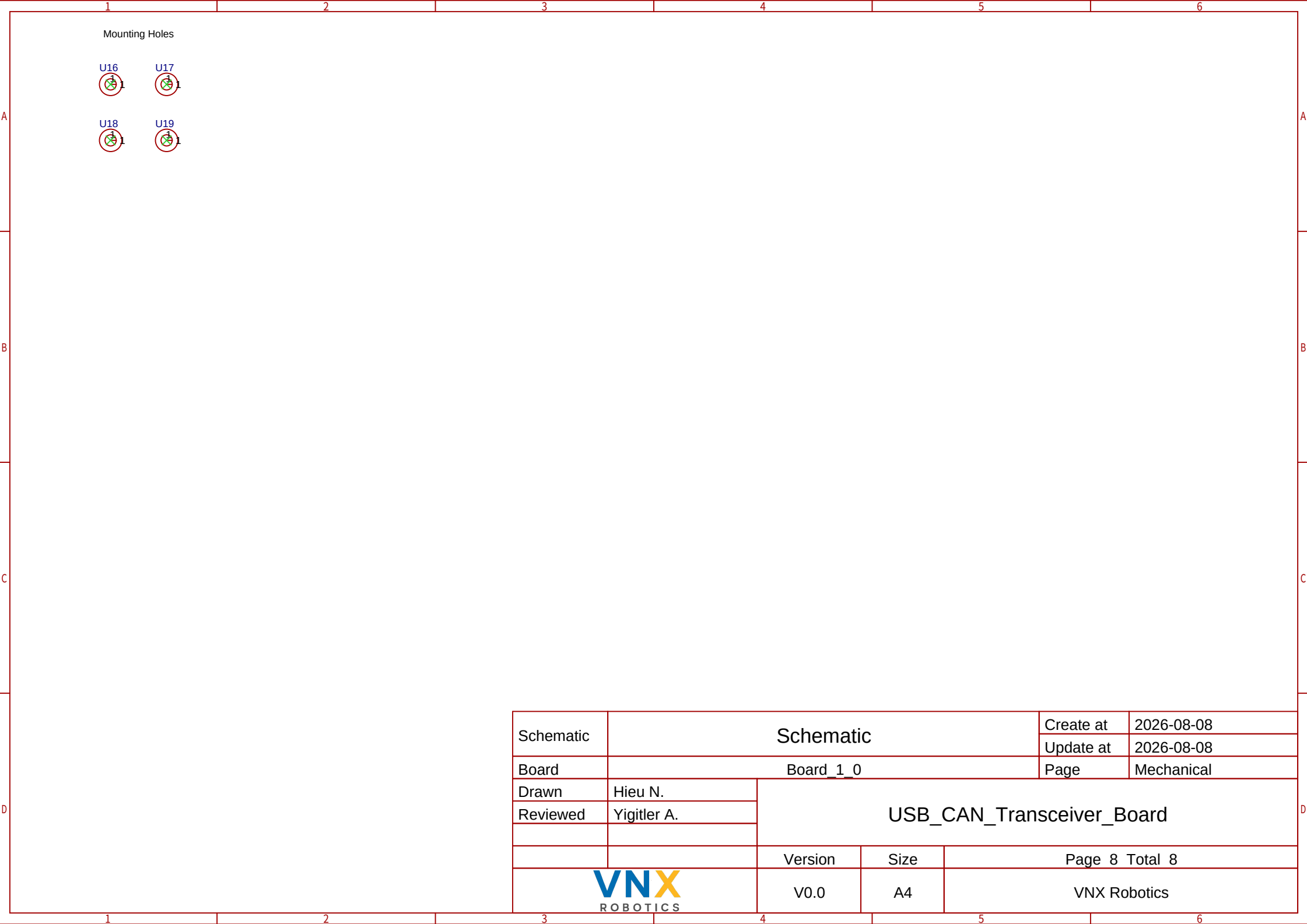
Schematic	Schematic		Create at	2026-08-05
			Update at	2026-08-09
Board	Board_1_0		Page	MCU_2
Drawn	Hieu N.	USB_CAN_Transceiver_Board		
Reviewed	Yigitler A.			
		Version	Size	Page 6 Total 8
VN X ROBOTICS		V0.0	A4	VNX Robotics



Specification / Parameter	TCAN3414DRBR	TCAN1044AVDRQ1
Main Supply Voltage (VCC)	3.0V to 3.6V (Nominal 3.3V)	4.5V to 5.5V (Nominal 5.0V)
Logic Level Shifting (VIO)	No (Logic referenced to VCC)	Yes (VIO supports 1.7V to 5.5V)
Supported Data Rates	Classical CAN & CAN FD up to 5 Mbps (Up to 8 Mbps in simple networks)	
Recessive Bus Bias Voltage	Nominal 1.9V	Nominal 2.5V (VCC / 2)
Operating Modes	Normal, Low-Power Standby, Ultra-Low Power Shutdown	Normal, Low-Power Standby (with Remote Wake WUP)
Application	Industrial	Automotive

Schematic		Schematic		Create at	2026-07-26
				Update at	2026-08-09
Board	Board_1_0			Page	CAN
Drawn	Hieu N.	USB_CAN_Transceiver_Board			
Reviewed	Yigitler A.				
		Version	Size	Page 7 Total 8	
		V0.0	A4	VNX Robotics	





Mounting Holes



Schematic	Schematic		Create at	2026-08-08
			Update at	2026-08-08
Board	Board_1_0		Page	Mechanical
Drawn	Hieu N.	USB_CAN_Transceiver_Board		
Reviewed	Yigitler A.			
		Version	Size	Page 8 Total 8
VNX ROBOTICS		V0.0	A4	VNX Robotics